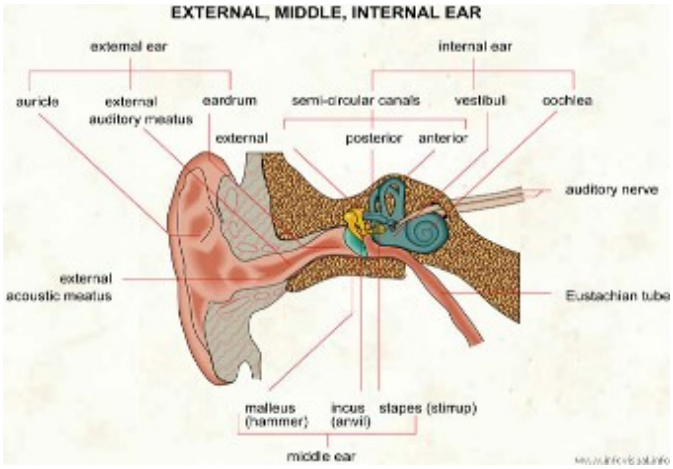
DEVELOPMENT OF EAR.

LEARNING OBJECTIVES.

- At the end of lecture, students should be able to know:

- Development of inner ear.
- Development of middle ear.
- Development of external ear.

EAR.

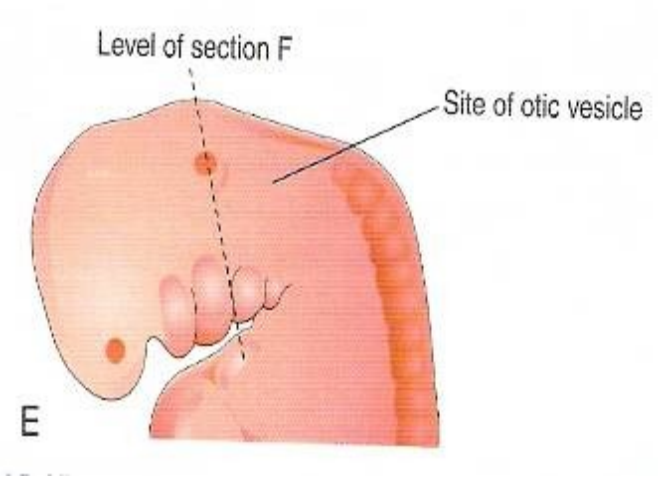


- Serves as an organ having both hearing and equilibrium function.

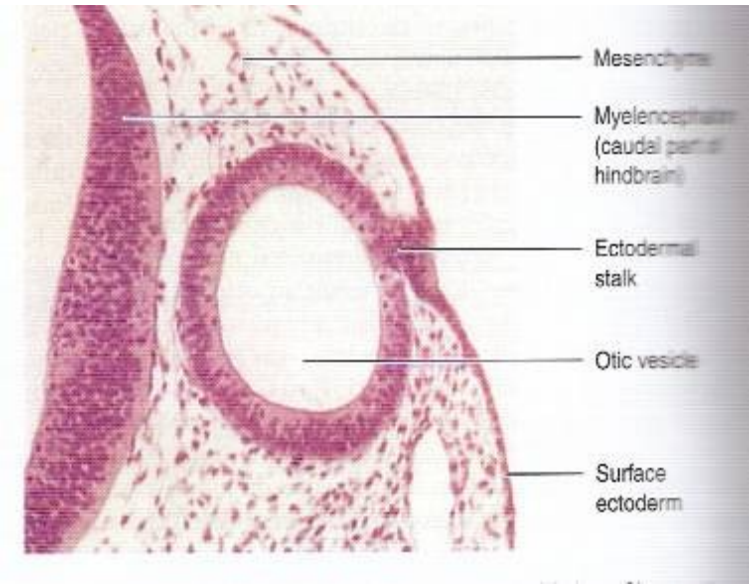
- Consists of 3 anatomical parts;

1. External Ear- consists of :
   - auricle (pinna).
   - external acoustic meatus.
   - external layer of eardrum.
2. Middle Ear- consists of :
   - three small auditory ossicles.
   - internal ear of tympanic membrane.
3. Internal Ear- consists of vestibulocochlear organ.

EAR.
DEVELOPMENT OF INNER EAR.



- It is first of three parts to develop.
- Early in fourth week, a thickening of surface ectoderm, OTIC PLACODE, appears on each side of myelencephalon.
- Each placode sinks deep into surface ectoderm into underlying mesenchyme results in otic pit.
- The edges of otic pits soon come together, fuse to form otic vesicle.
- With further development each vesicles divides into MEMBRANOUS LABYRINTH.



Development of membranous labyrinth.

- The otic vesicle, precursor of the **definitive membranous labyrinth**, differentiates into 3 parts:

1. A dorsomedial, elongated **endolymphatic appendage**, which will form:

- The **endolymphatic duct.**
- The **endolymphatic sac** at the distal end of the **endolymphatic duct.**

Development of membranous labyrinth.

2. A central, expanded **utricular region**, which will form:

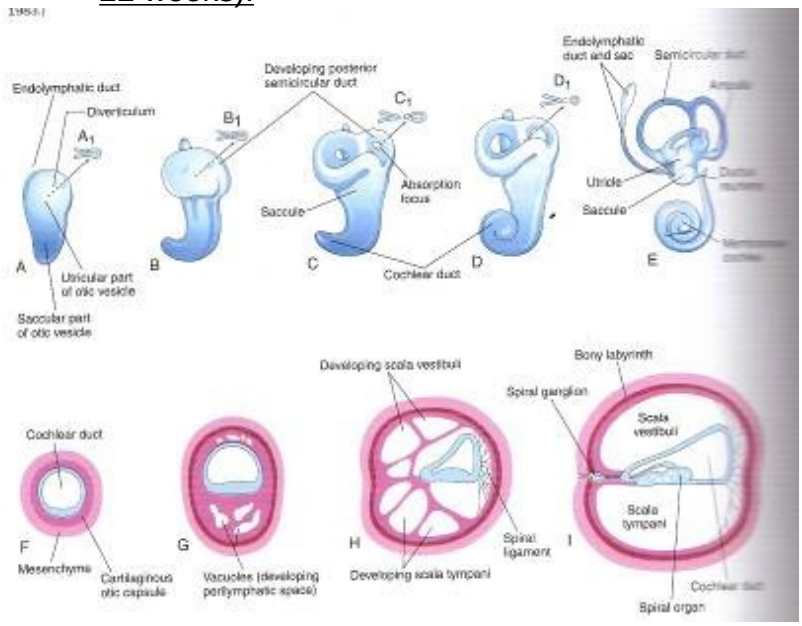
- The **utricle.**
- The **3 semicircular ducts**, which arise from **utricular diverticula** that appear in the 7th week.

3. A ventral, conical **saccular region**, which forms:

- The **saccule**
- The **cochlear duct**, which coils to about **2.5 spirals** and elongates in the 5th week, with the spiral **organ of Corti** differentiating in the 7th week.
- The **ductus reuniens** joining the saccule and cochlear duct

▶ The cartilagenous otic capsule latter ossifies to form <u>BONY LABYRINTH</u> of internal ear.

▶ The internal ear reaches, its adult size and shape by middle of fetal period <u>(20-22 weeks).</u>

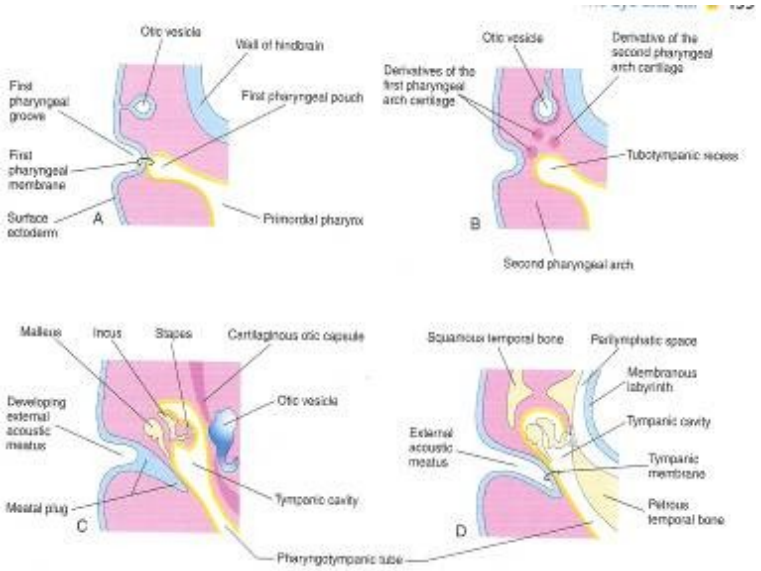


<u>THE DEVELOPMENT OF MIDDLE EAR.</u>

➢ The tympanic cavity is of <u>endodermal origin</u> and is derived from <u>1st pharayngeal pouch</u>.

➢ The proximal part of tubotympanic recess forms the <u>PHARYNGO-TYMPANIC TUBE OR AUDITORY TUBE</u> connect with pharynx.

➢ The distal part of recess expands & become <u>tympanic cavity.</u>



DEVELOPMENT OF OSSICLES.

- Malleous, incus are derived from cartilage of 1st pharyngeal arch.
- Stapes develop from 2nd pharyngeal arch.
- Ossicles appear during first half of fetal life.
- The endodermal epithelial lining of primitive tympanic cavity expands in the space & become twice as large as before.
- During late fetal period expansion of tympanic cavity gives rise to mastoid antrum, located in the petro mastoid part of temporal bone.

Development of middle ear.

Development of Middle Ear.

DEVELOPMENT OF EXTERNAL EAR.

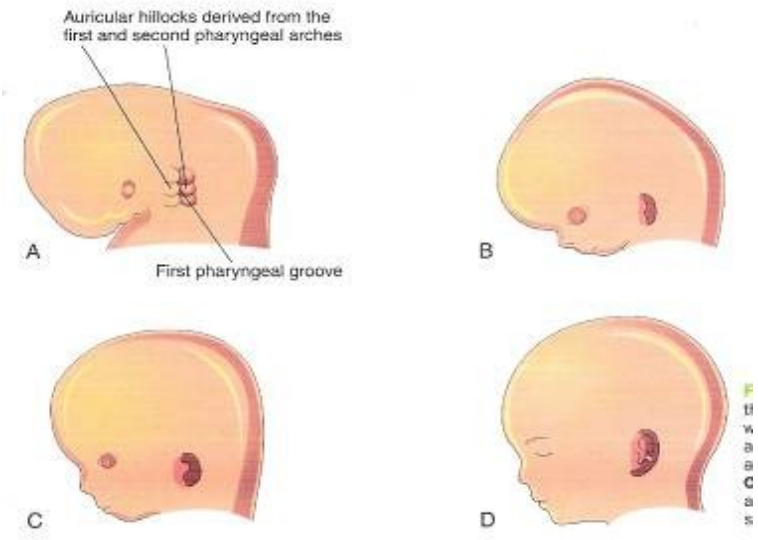
Pharyngeal arch derivatives in the middle ear	
Pharyngeal arch	**Middle ear structures**
1st	• Cartilage ---> malleus, incus • Mesoderm ---> tensor tympani
2nd	• Cartilage ---> stapes • Mesoderm ---> stapedius

- The external acoustic meatus develops from dorsal part of first pharyngeal cleft.

- The ectodermal epithelium of first pharyngeal cleft forms external layer of tympanic membrane the internal membrane is formed by endodermal epithelium.

- The auricle develops from 3 pairs of **auricular hillocks** that arise in the 5th week, on the lateral faces of the 1st and 2nd pharyngeal arches.

- In the **7th week**, the auricular hillocks begin to enlarge, differentiate and fuse, producing the final shape of the ear.



CLINICAL CORRELATION.

MICROTIA.



- Microtia is a condition in which the external portion of the ear (the auricle) is malformed.

ANOTIA.



- Anotia is a very rare congenital anomaly in which the external part of the ear is entirely absent.

SELF ASSESSMENT.

- The otic vesicle differentiates into which three parts?

- Malleus, incus are derived from which pharyngeal arch?
- First pharyngeal arch gives rise to which part of auricle?

THANK YOU ALL■